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University AI Copilot:
Intelligent Support for Teaching, Services, and Decision-Making

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Use Cases Received

- AI chatbot for student services and internal knowledge routing
- AI feedback analysis of student satisfaction comments
- AI review of student CVs and application documents
- AI support for checking freelancer documentation
- AI-based market research and competitor monitoring
- AI trend anaylsis for innovation management
- CORE-Empowerment Bot for teaching support

Challenge

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Universities face increasing complexity in:

- teaching innovation and curriculum development
- student services and communication
- administrative processes
- strategic decision-making

Manual knowledge work → information scattered → AI can automate analysis, routing, summarization, and decision support.

Core Idea

- The project explores how **AI agents can support different functions in higher education**.
- The goal is to design a **modular AI platform** that assists:
  - Lecturers
  - Administrative staff
  - University management
  - Students
- The system provides **knowledge retrieval, analysis, and decision support** while keeping human expertise at the center.

Modular AI-Agent Architecture

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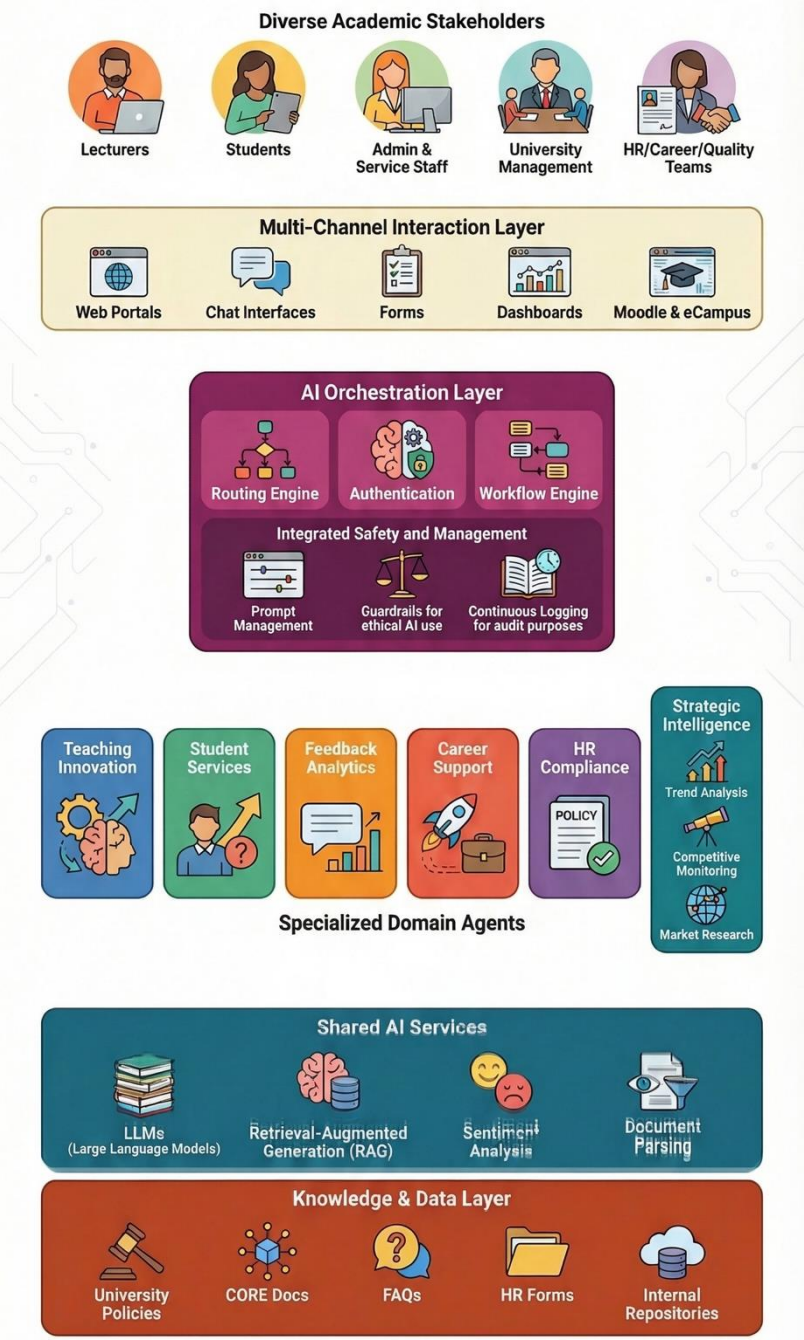
- Instead of developing isolated tools, the project proposes a **multi-agent AI system for universities**.
- University AI Copilot
  - Teaching Innovation Agent
  - Student Service Agent
  - Feedback Analytics Agent
  - Career Support Agent
  - HR Compliance Agent
  - Strategic Intelligence Agent
  - International Office Agent


Diverse Academic Stakeholders
Lecturers
Students
Admin & Service Staff
University Management
HR/Career/Quality Teams
Multi-Channel Interaction Layer
Web Portals
Chat Interfaces
Forms
Dashboards
Moodle & eCampus
AI Orchestration Layer
Routing Engine
Authentication
Workflow Engine
Integrated Safety and Management
Prompt Management
Guardrails for ethical AI use
Continuous Logging for audit purposes
Teaching Innovation
Student Services
Feedback Analytics
Career Support
HR Compliance
POLICY
Strategic Intelligence
Trend Analysis
Competitive Monitoring
Market Research
Specialized Domain Agents
Shared AI Services
LLMs (Large Language Models)
Retrieval-Augmented Generation (RAG)
Sentiment Analysis
Document Parsing
Knowledge & Data Layer
University Policies
CORE Docs
FAQs
HR Forms
Internal Repositories

Teaching Innovation Agent (CORE Bot)

- Supports lecturers in designing competence-oriented courses.
- Functions
  - Generative learning objectives
  - support course and module design
  - suggest teaching methods
  - design assessments and rubrics
  - integrate AI into teaching responsibly

Goal:
Support lecturers in implementing **competence-oriented teaching frameworks.**

Student Services Chatbot

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- Supports students in finding information and navigating university services.
- Functions
  - answer frequently asked questions
  - identify student needs
  - route requests to the correct department
  - provide links to relevant services

AI Based Feedback Analysis

- Analyzes large volumes of student feedback and survey comments.
- Functions
  - automatic clustering of comments
  - sentiment analysis
  - detection of recurring issues
  - generation of summary reports

Career Support AI

- Helps students improve their job application documents.
- Functions:
  - analyze CVs and cover letters
  - Detect common errors
  - suggest improvements
  - prepare students before career consultations.

HR Compliance Agent

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- Supports HR in reviewing documentation of external lecturers and freelancers.
- Functions
  - analyze CVs and documentation
  - detect missing information
  - identify potential risk indicators.

Strategic Intelligence Agent

- Supports university leadership with **market and trend analysis**.
- Functions
  - competitor monitoring
  - trend detection in higher education
  - automated reports and dashboards.

Expected Outcomes

- design a modular AI-agent architecture
- develop **one prototype AI assistant**
- evaluate the solution with users
- present recommendations for implementation.

Thank you for joining us!

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